

Operating System Tutorial for Lab Assignment-2

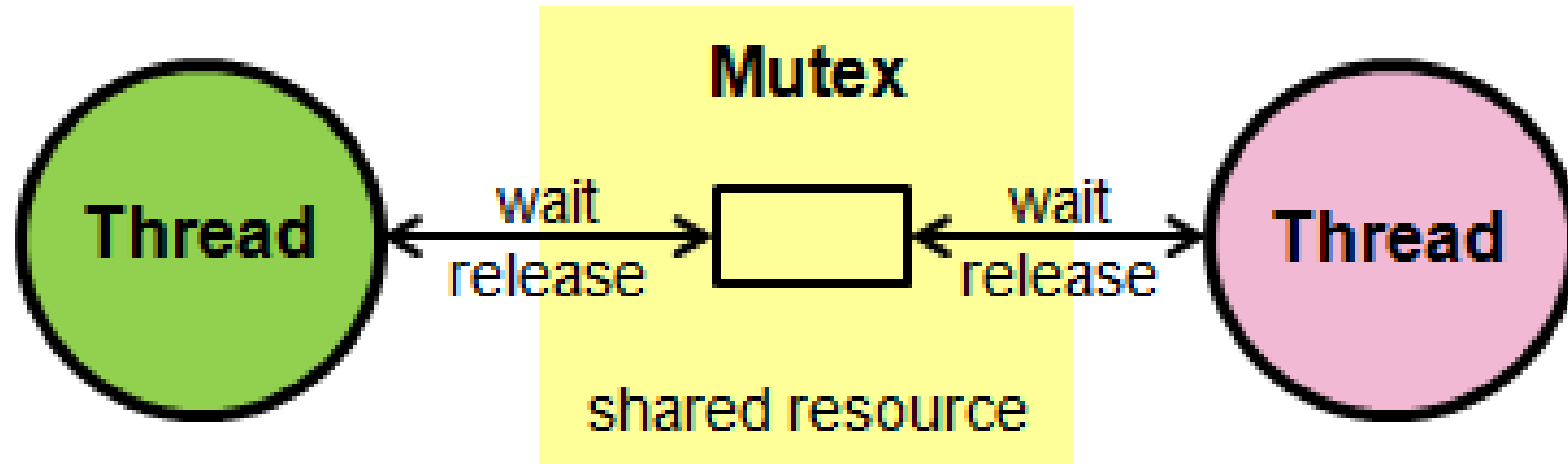
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Mutex

- Short for “Mutual Exclusion”
- In computer programming, a mutex is a Program object that allows a thread to share the same resource, such as file access, but not simultaneously.
- When a program is started, a mutex is created with a unique name. After this stage, any thread that needs the resource must lock the mutex from other threads while it is using the resource. The mutex is set to unlock when the data is no longer needed or the routine is finished.
- ex) Changing Room (Only one stall)

Mutex (Continued)

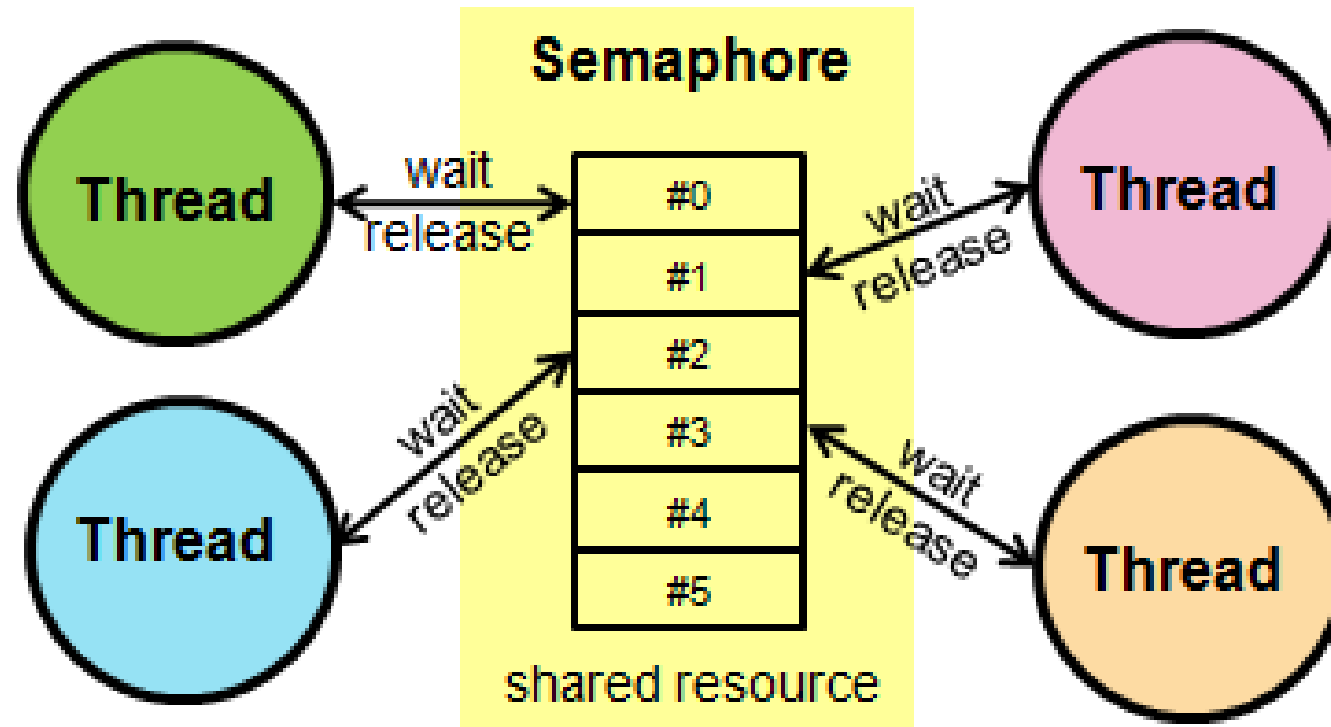


Only one thread can have access to the shared resource at a time

Semaphores

- A hardware or software flag. In Multitasking systems, a semaphore is a variable with a value that indicates the status of a common resource. It's used to lock the resource that is being used. A process needing the resource checks the semaphore to determine the resource's status and then decides how to proceed.
- Ex) Changing Room (Multiple Stalls)

Semaphores (Continued)



Threads share access to a resource. The Semaphore only controls the number of threads with access

Common Issues

- Requesting a resource and forgetting to release it;
- Releasing a resource that was never requested;
- Holding a resource for a long time without needing it;
- Using a resource without requesting it first (or after releasing it).

Programming Example